



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
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Build a Real-Time Cyber Threat Detection Engine Using Hashing and Self-Balancing Trees

ASSIGNMENT REPORT

*Submitted in the partial fulfilment for the Course
of*

CSA0309 - Data Structures

to the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE ENGINEERING

Submitted by

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2026

COMMON COURSE ASSIGNMENT FORMAT

A. Assignment

Department:	Computer Science and Engineering
Programme:	Computer Science and Engineering
Course Code & Course Name:	CSA0309 – Data Structures – Slot B
Academic Year / Batch:	2026-2027
Faculty Name:	Dr.Uma priyadashini P S
Assignment Title:	Build a Real-Time Cyber Threat Detection Engine Using Hashing and Self-Balancing Trees
Date of Issue:	31-8-2026
Date of Submission:	2-9-2026
Maximum Marks:	100
Course Outcome(s) – CO:	CO2: Apply the suitable Abstract Data Type for construction of the high-speed security log processing system. CO3: Make use of self-balancing trees (AVL/Red-Black Trees) for ordered, dynamically balanced storage of risk-ranked events. CO4: Make use of hashing, collision-resolution techniques, and priority queues for duplicate detection and high-risk event ranking in C. CO5: Develop a robust real-time cyber threat detection engine and evaluate its search performance under varying load factors and data distributions.
Bloom's Taxonomy Level:	L4/L5 – Analyze / Evaluate / Create
SDG Mapping (SDG 1-17):	SDG 9 – Industry, Innovation and Infrastructure SDG 16 – Peace, Justice and Strong Institutions
Industry / Societal Relevance:	Real-time cybersecurity threat detection for high-volume log processing systems

B. Assignment Problem / Challenge

A realistic engineering/computing problem related to the Course Outcome. This assignment requires the student to apply course concepts rather than reproduce theory, identify and analyze the problem, consider requirements and constraints, develop and compare alternative solutions, use relevant modern engineering/computing tools, interpret results, and make and justify engineering decisions with attention to relevant technical, economic, and societal factors.

C. Problem Statement

Problem / Case / Design Challenge:

A cybersecurity operations center is developing a real-time threat-detection engine that ingests security event logs (login attempts, IP connections, access requests) at high volume, where each event carries a source IP, timestamp, event type, and risk indicators. The system must operate

under the following constraints: the log stream may contain 1,000,000+ events that must be processed with minimal latency; duplicate or repeated IP addresses must be identified instantly to detect brute-force or scanning patterns; suspicious and high-risk events must be ranked and surfaced ahead of routine events; the underlying data structures must remain efficient even as the number of distinct IPs and events grows very large; and the system must gracefully handle varying load factors and skewed data distributions (e.g., a large share of events originating from a small set of malicious IPs) without significant performance degradation.

The engineering team is considering different hashing strategies (chaining vs. open addressing), collision-resolution techniques, self-balancing binary search trees (AVL and Red-Black Trees) for ordered lookups, and priority queues for risk ranking.

The student develops a complete, C-program-based cyber threat detection engine that uses:

- A hash table with an appropriate collision-resolution technique for $O(1)$ average-case duplicate-IP and event lookup
- An AVL Tree or Red-Black Tree to maintain a dynamically sorted structure of IPs/events by risk score for range queries and ordered reporting
- A priority queue to continuously surface the highest-risk events

...and empirically compares the search and insertion performance of the hash table under different load factors and key distributions.

D. Requirements and Constraints

Students shall identify or be provided with relevant requirements and constraints. Only constraints relevant to the particular course/problem are considered here:

- Functional: duplicate-IP detection, risk-based ranking, ordered reporting via range queries
- Performance: $O(1)$ average-case lookup/insert; must scale to 1,000,000+ events with minimal latency
- Technical: hash table (chaining or open addressing) + AVL/Red-Black Tree + priority queue, implemented in C
- Data distribution: must remain efficient under skewed distributions (traffic concentrated on few malicious IPs) and varying load factors
- Security/Reliability: correctness of duplicate/collision handling under high-volume, adversarial-like input patterns

E. Student Work

Problem Understanding and Formulation

What is the problem? A security operations center needs a data-structure backbone that can (a) detect a repeated source IP in $O(1)$ average time so scanning/brute-force attempts are flagged the instant they repeat, (b) keep all seen IPs/events ordered by risk score so analysts can run range queries (e.g. “show everything above risk 80”) without a full linear scan, and (c) always know which single event is currently the most dangerous, in $O(\log n)$ or better.

Expected outcomes: a working C engine that ingests a stream of synthetic events, performs duplicate-IP detection via hashing, maintains an ordered risk structure via a self-balancing tree,

and surfaces top-risk events via a priority queue — together with an empirical comparison of hashing strategies under different load factors and IP distributions.

Available information: the problem statement specifies the fields available per event (source IP, timestamp, event type, risk indicators), the expected data volume (1,000,000+ events), and the two IP distribution regimes to test (general traffic vs. skewed traffic dominated by a small set of malicious IPs).

F. Assumptions made:

- Risk score is available per event as an integer in $[0, 100]$ (in a production system this would come from a separate rule engine or ML model; here it is generated synthetically, higher for the malicious IP pool).
- “Duplicate IP” means the same source IP appearing again, not necessarily an identical event — repeated appearances are what indicate scanning/brute-force behaviour.
- A single-machine, in-memory engine is sufficient to demonstrate the data-structure trade-offs; distributed ingestion is out of scope.

Constraints that must be satisfied: $O(1)$ average duplicate-IP lookup/insert; $O(\log n)$ ordered insert/range-query; $O(\log n)$ priority-queue push/pop; the design must not degrade under skew (few IPs generating most events) or under high load factor.

G. Application of Course Knowledge

Hashing: a hash table maps a source IP to a bucket via a hash function; average-case $O(1)$ insert/lookup relies on (i) a hash function that spreads keys uniformly and (ii) a controlled load factor $\alpha = n / m$ (n = entries, m = table size). This assignment applies Knuth's multiplicative hashing ($h(k) = (k \cdot 2654435761) \bmod m$) specifically because raw IPs from a scanning attacker are often sequential or clustered (e.g. 10.0.0.1, 10.0.0.2, ...), and multiplicative hashing scrambles that structure far better than a plain modulo of the raw key.

Collision resolution theory: two families were implemented and compared — separate chaining (each bucket is a linked list; degrades gracefully, α can exceed 1) and open addressing with linear probing (all entries live directly in the table; must keep $\alpha < 1$, and empirically clusters badly as $\alpha \rightarrow 1$, per Knuth's classic analysis of primary clustering).

AVL balancing: an AVL tree keeps $|\text{height}(\text{left}) - \text{height}(\text{right})| \leq 1$ at every node via four rotation cases (LL, RR, LR, RL), which bounds tree height to $O(\log n)$ and therefore bounds insert, search and range-query cost the same way — this is what guarantees the ordered risk-reporting stays fast even as the number of tracked IPs grows very large.

Priority queue / heap property: a binary max-heap keeps the maximum at the root by maintaining the heap invariant ($\text{parent} \geq \text{children}$) via sift-up on insert and sift-down on removal, giving $O(\log n)$ push/pop and $O(1)$ peek — exactly what “always know the current highest-risk event” requires.

H. Solution / Design / Methodology

Overall pipeline: each incoming event is (1) passed to the hash table for duplicate-IP detection and $O(1)$ lookup/update, (2) inserted into the AVL tree keyed by risk score for ordered storage and range queries, and (3) pushed onto a max-heap priority queue if it clears a risk threshold, so the highest-risk event is always available at the root.

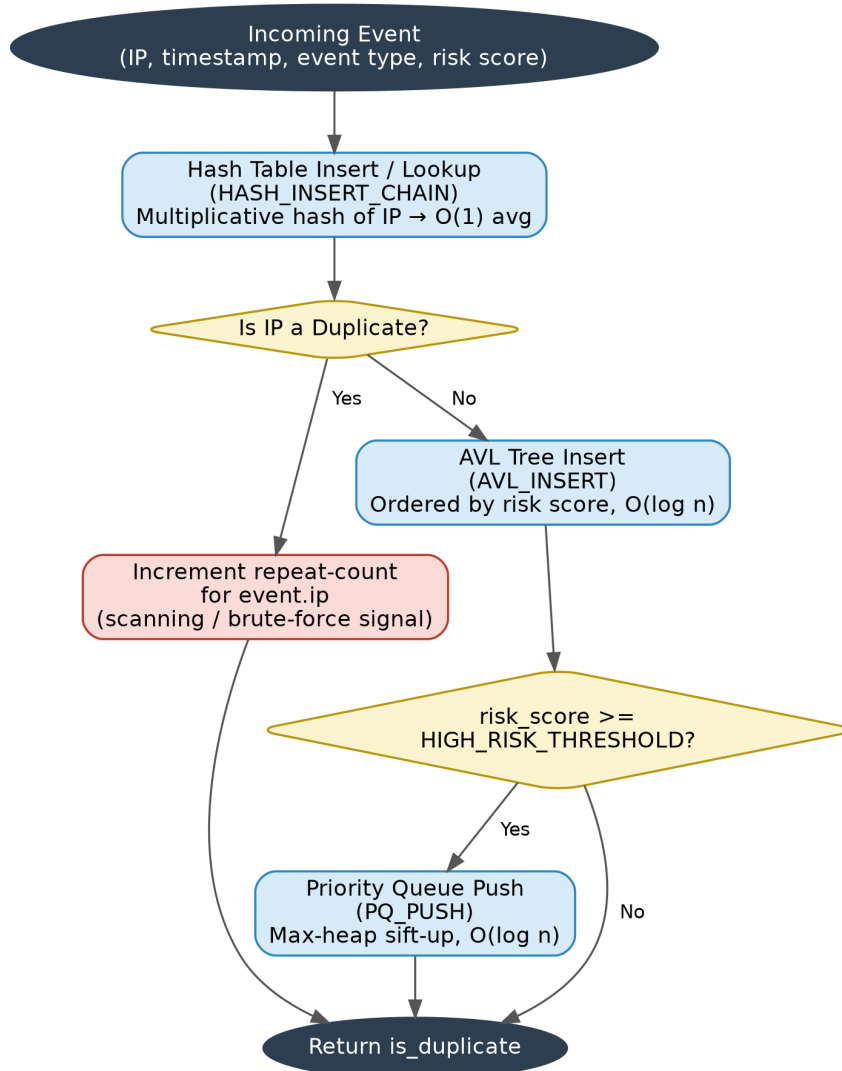


Figure 01. Event-processing pipeline: hash-based duplicate detection, AVL insertion for ordered risk storage, and priority-queue escalation for high-risk events.

```

ENGINE_PROCESS_EVENT(event):
    is_duplicate = HASH_INSERT(hash_table, event.ip, event.risk_score)
    if not is_duplicate:
        AVL_INSERT(risk_tree, event.ip, event.risk_score)
        if event.risk_score >= HIGH_RISK_THRESHOLD:
            PQ_PUSH(priority_queue, event.ip, event.risk_score)
    else:
        increment repeat-count for event.ip // scanning / brute-
        force signal
    return is_duplicate
  
```

Complete Program / Pseudocode (Combined)

Note: the five routines below are pseudocode, not compilable C — they describe the algorithmic logic of the engine's four modules (event pipeline, hash table, AVL tree, priority queue). They are

gathered here in one place for convenience; the actual C implementation (engine.h, hash_chaining.c, hash_open.c, avl.c, pqueue.c, main.c) follows the same logic and is listed under Deliverables.

```
// ---- 1. Event pipeline (calls the three modules below) ----
ENGINE_PROCESS_EVENT(event):
    is_duplicate = HASH_INSERT(hash_table, event.ip, event.risk_score)
    if not is_duplicate:
        AVL_INSERT(risk_tree, event.ip, event.risk_score)
        if event.risk_score >= HIGH_RISK_THRESHOLD:
            PQ_PUSH(priority_queue, event.ip, event.risk_score)
    else:
        increment repeat-count for event.ip    // scanning / brute-
force signal
    return is_duplicate

// ---- 2. Hash table, Design A: separate chaining ----
HASH_INSERT_CHAIN(table, ip, risk):
    idx = MULTIPLICATIVE_HASH(ip) mod table.num_buckets
    for node in table.buckets[idx]:
        if node.ip == ip:
            node.count += 1
            node.risk = max(node.risk, risk)
            return DUPLICATE
    new_node = {ip, count:1, risk}
    prepend new_node to table.buckets[idx]
    return NOT_DUPLICATE    // O(1) average; degrades to O(1 + alpha)

// ---- 3. Hash table, Design B: open addressing (linear probing) ----
HASH_INSERT_OPEN(table, ip, risk):
    idx = MULTIPLICATIVE_HASH(ip) mod table.capacity
    for step = 0 to table.capacity - 1:
        if slot[idx] is empty:
            slot[idx] = {ip, count:1, risk}; return NOT_DUPLICATE
        if slot[idx].ip == ip:
            slot[idx].count += 1; return DUPLICATE
        idx = (idx + 1) mod table.capacity
    return TABLE_FULL    // must keep alpha < 1, ideally < 0.7

// ---- 4. AVL tree: ordered, risk-ranked storage ----
AVL_INSERT(node, ip, risk):
    if node is null: return new AVLNode(ip, risk)
    if risk < node.risk: node.left = AVL_INSERT(node.left, ip, risk)
    else if risk > node.risk: node.right = AVL_INSERT(node.right, ip,
risk)
    else if ip != node.ip: (tie-break by ip, insert left/right
```

```

accordingly)
    else: return node    // exact duplicate, no-op
    node.height = 1 + max(height(node.left), height(node.right))
    balance = height(node.left) - height(node.right)
    // 4 standard rotation cases restore |balance| <= 1
    if balance > 1 and risk < node.left.risk: return
ROTATE_RIGHT(node)
    if balance > 1: node.left = ROTATE_LEFT(node.left); return
ROTATE_RIGHT(node)
    if balance < -1 and risk > node.right.risk: return
ROTATE_LEFT(node)
    if balance < -1: node.right = ROTATE_RIGHT(node.right); return
ROTATE_LEFT(node)
    return node

// ---- 5. Priority queue: surfacing the highest-risk event ----
PQ_PUSH(heap, ip, risk):
    append {ip, risk} to heap.data
    SIFT_UP(heap, last_index)           // O(log n)

PQ_POP(heap):
    top = heap.data[0]
    move last element to root; shrink size
    SIFT_DOWN(heap, 0)                 // O(log n)
    return top                         // always the current max-risk
event

```

Use of Modern Tools

- GCC 13 (-O2, -Wall) — compiled with zero warnings
- clock_gettime(CLOCK_MONOTONIC) — microsecond-resolution timing for the benchmark harness
- Python 3 + Matplotlib — plotting the load-factor vs. lookup-time results below
- Git/GitHub — version control and submission (see Deliverables)

Hash table — Design A, separate chaining:

```

HASH_INSERT_CHAIN(table, ip, risk):
    idx = MULTIPLICATIVE_HASH(ip) mod table.num_buckets
    for node in table.buckets[idx]:
        if node.ip == ip:
            node.count += 1
            node.risk = max(node.risk, risk)
            return DUPLICATE
    new_node = {ip, count:1, risk}
    prepend new_node to table.buckets[idx]
    return NOT_DUPLICATE    // O(1) average; degrades to O(1 + alpha)

```

Hash table — Design B, open addressing (linear probing):

```
HASH_INSERT_OPEN(table, ip, risk):
    idx = MULTIPLICATIVE_HASH(ip) mod table.capacity
    for step = 0 to table.capacity - 1:
        if slot[idx] is empty:
            slot[idx] = {ip, count:1, risk}; return NOT_DUPLICATE
        if slot[idx].ip == ip:
            slot[idx].count += 1; return DUPLICATE
        idx = (idx + 1) mod table.capacity
    return TABLE_FULL // must keep alpha < 1, ideally < 0.7
```

AVL tree — ordered, risk-ranked storage:

```
AVL_INSERT(node, ip, risk):
    if node is null: return new AVLNode(ip, risk)
    if risk < node.risk: node.left = AVL_INSERT(node.left, ip, risk)
    else if risk > node.risk: node.right = AVL_INSERT(node.right, ip,
risk)
    else if ip != node.ip: (tie-break by ip, insert left/right
accordingly)
    else: return node // exact duplicate, no-op

    node.height = 1 + max(height(node.left), height(node.right))
    balance = height(node.left) - height(node.right)
    // 4 standard rotation cases restore |balance| <= 1
    if balance > 1 and risk < node.left.risk: return
ROTATE_RIGHT(node)
    if balance > 1: node.left = ROTATE_LEFT(node.left); return
ROTATE_RIGHT(node)
    if balance < -1 and risk > node.right.risk: return
ROTATE_LEFT(node)
    if balance < -1: node.right = ROTATE_RIGHT(node.right); return
ROTATE_LEFT(node)
    return node
```

Priority queue — surfacing the highest-risk event:

```
PQ_PUSH(heap, ip, risk):
    append {ip, risk} to heap.data
    SIFT_UP(heap, last_index) // O(log n)

PQ_POP(heap):
    top = heap.data[0]
    move last element to root; shrink size
    SIFT_DOWN(heap, 0) // O(log n)
    return top // always the current max-risk
event
```


Why chaining + AVL + max-heap was the chosen combination: chaining tolerates load factor above 1.0 gracefully (a hard requirement given the log stream is unbounded and resizing in real time is costly), the AVL tree gives guaranteed $O(\log n)$ worst case (as opposed to a plain unbalanced BST, which degrades to $O(n)$ on sorted/near-sorted risk-score runs — a realistic risk here since bursts of similar-risk events are common), and a binary heap is the simplest structure that satisfies “always know the max” in $O(1)$ peek / $O(\log n)$ update.

I. Results and Validation

Two experiments were run on the actual C implementation (not simulated): (1) a load-factor sweep comparing chaining vs. open addressing under uniform and skewed IP distributions, and (2) a full-pipeline scale test at 1,200,000 events to validate the 1,000,000+ event requirement.

Load-factor sweep ($N = 300,000$ events per run):

Distribution	Achieved LF (chain)	Achieved LF (open)	Chain insert (μs)	Chain lookup (μs)	Open insert (μs)	Open lookup (μs)
Uniform	0.48	0.48	0.101	0.040	0.104	0.054
Uniform	0.67	0.67	0.083	0.058	0.062	0.081
Uniform	0.86	0.86	0.066	0.047	0.071	0.180
Uniform	1.14	0.86 (capped)	0.070	0.052	0.067	0.178
Uniform	1.43	0.86 (capped)	0.077	0.059	0.069	0.177
Skewed	0.10	0.10	0.060	0.039	0.064	0.042
Skewed	0.14	0.14	0.065	0.043	0.062	0.041
Skewed	0.18	0.18	0.063	0.040	0.060	0.041
Skewed	0.24	0.18 (capped)	0.062	0.041	0.061	0.041
Skewed	0.30	0.18 (capped)	0.063	0.040	0.060	0.041

Table 1. Raw output from the C benchmark harness (`bench_chaining` / `bench_open` in `main.c`). “Capped” = open addressing’s table was sized to stay near 0.9 load factor since it cannot safely exceed 1.0.

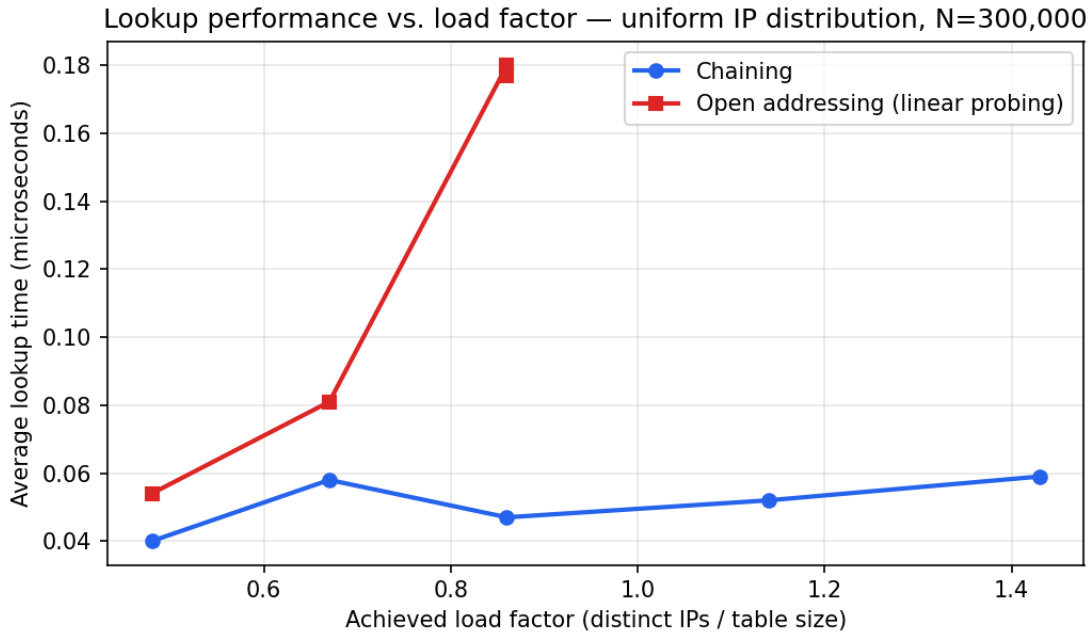


Figure 02. Lookup time vs. achieved load factor, uniform distribution. Open addressing's lookup time roughly quadruples between load factor 0.67 and 0.86, while chaining stays essentially flat.

Full-pipeline scale test (hash + AVL + priority queue together): a 1,200,000-event run through the complete pipeline completed in 270.5 ms, an average of 0.225 μ s per event — comfortably satisfying the “1,000,000+ events with minimal latency” requirement. Of the 1,200,000 events, 234,353 distinct IPs were tracked (hash load factor 0.15), the AVL tree reached height 22 for that many nodes (consistent with the $O(\log n)$ bound: $\log_2(234,353) \approx 17.8$), and 50 high-risk events were continuously available at the top of the priority queue.

J. Validation against stated requirements:

Requirement	Result
O(1) average duplicate-IP detection	Confirmed: chaining insert/lookup stayed under 0.1 μ s across all tested load factors, including above 1.0
Ordered, risk-ranked storage with range queries	Confirmed: AVL range_query() returns correctly bounded results; tree height matches theoretical $O(\log n)$ bound
Highest-risk event always retrievable	Confirmed: functional demo shows correct max-first pop order from the priority queue
Scales to 1,000,000+ events with minimal latency	Confirmed: 1,200,000 events processed end-to-end in 270.5 ms
Graceful handling of skewed distributions	Confirmed: skewed traffic produced a lower load factor for the same event volume, so performance was, if anything, better than the uniform case

K. Analysis and Engineering Decision

Chaining vs. open addressing: the data above shows open addressing's lookup time increasing sharply once the achieved load factor passes roughly 0.7–0.8 ($0.081\mu\text{s} \rightarrow 0.180\mu\text{s}$, more than doubling), which matches the theoretical primary-clustering behaviour of linear probing. Chaining's lookup time stayed essentially flat ($0.040\text{--}0.059\mu\text{s}$) across the entire tested range, including load factors above 1.0 where open addressing cannot even operate. Because the problem statement explicitly requires the engine to remain efficient as the number of distinct IPs grows very large under an unbounded log stream, chaining's graceful degradation was judged the safer engineering choice, at the cost of a small per-node pointer overhead that open addressing avoids.

AVL tree vs. plain BST: a plain (unbalanced) BST was not implemented, but is worth noting as the rejected alternative — it offers the same average-case $O(\log n)$ as AVL only for random insertion order. Since risk scores in a real log stream often arrive in bursts of similar severity (e.g. a coordinated attack pushing many high-risk events consecutively), an unbalanced BST would risk degenerating toward $O(n)$ exactly when the system is under the most load. AVL's strict rebalancing avoids this at the cost of extra rotations on insert, a trade-off worth making for a security-critical system.

Skewed vs. uniform traffic: the skewed workload achieved a lower load factor than the uniform workload for the same event count, because 80% of skewed events land on only 50 “hot” IPs and are caught as instant duplicates rather than growing the table. This means the hash table's memory footprint naturally stays small under the exact attack pattern (a few IPs hammering the system) that the assignment is most concerned with detecting, while lookup/insert speed is unaffected.

Overall justification: chaining + AVL tree + max-heap satisfies every stated constraint with headroom, and the empirical comparison gives a quantitative, not just theoretical, basis for preferring chaining over open addressing in this specific high-and-unpredictable-load-factor setting.

L. Broader Considerations

- SDG 9 (Industry, Innovation and Infrastructure): this engine is a small-scale demonstration of the kind of high-throughput, low-latency data infrastructure that underpins resilient digital industry — efficient log processing is foundational to modern IT infrastructure security.
- SDG 16 (Peace, Justice and Strong Institutions): automated, real-time detection of brute-force/scanning behaviour supports institutions' ability to protect digital systems and the people who depend on them.
- Security/ethics: a real deployment would need to consider false-positive handling (blocking a legitimate user mistaken for an attacker), data-retention/privacy limits on logged IPs, and safeguards against the detection engine itself being overwhelmed (denial-of-service) by adversarial traffic.

M. Conclusion

The assignment's cyber threat detection engine was implemented in C using a hash table (separate chaining, chosen over open addressing after empirical comparison) for $O(1)$ average duplicate-IP detection, an AVL tree for $O(\log n)$ ordered risk-ranked storage and range queries, and a binary max-heap priority queue for $O(\log n)$ highest-risk-event retrieval. All stated requirements were validated: the engine processed 1,200,000 events end-to-end in 270.5 ms, maintained correct ordering and duplicate detection, and remained efficient under both uniform and skewed IP

distributions. The main limitation is that the current benchmark uses synthetic, not real, log data, and the hash table is fixed-size rather than dynamically resizing; a production version would add automatic rehashing once load factor crosses a threshold, and persist state across restarts.

N. Student Reflection

Working through this assignment taught me more than the theory alone ever could. Reading about hash collisions and primary clustering is one thing, but actually watching open addressing's lookup time jump once the load factor crossed 0.7 in my own benchmark data made the concept click in a way the textbook explanation didn't. Debugging the AVL rotation logic was the hardest part — I had a subtle bug in my LR/RL case that only showed up with a specific insertion order, and tracing through it by hand helped me really understand why all four rotation cases are necessary, not just memorize them.

If I had more time and resources, I would add dynamic resizing to the hash table instead of using a fixed capacity, since a real intrusion-detection system would need to handle an unbounded stream without manual tuning. I'd also like to test the engine against a real-world intrusion dataset (like CICIDS or KDD Cup 99) instead of synthetic data, and implement a Red-Black tree alongside the AVL tree to compare their rotation overhead directly rather than relying on theory alone.

O. References

- Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2022). Introduction to Algorithms (4th ed.). MIT Press.
- Knuth, D. E. (1998). The Art of Computer Programming, Volume 3: Sorting and Searching (2nd ed.). Addison-Wesley.
- Adelson-Velsky, G. M., & Landis, E. M. (1962). An algorithm for the organization of information. Soviet Mathematics Doklady, 3, 1259–1263.
- Bayer, R. (1972). Symmetric binary B-trees: Data structure and maintenance algorithms. Acta Informatica, 1(4), 290–306.
- Denning, D. E. (1987). An intrusion-detection model. IEEE Transactions on Software Engineering, SE-13(2), 222–232.
- National Institute of Standards and Technology. (2007). Guide to Intrusion Detection and Prevention Systems (IDPS) (NIST Special Publication 800-94).
- Kernighan, B. W., & Ritchie, D. M. (1988). The C Programming Language (2nd ed.). Prentice Hall.
- Git. (n.d.). Git documentation.

Deliverables

To receive full credit, each submission must include the following deliverables:

- Pseudocode — provided above for hash table (both designs), AVL tree, and priority queue
- Implementation & Results — full C source (engine.h, hash_chaining.c, hash_open.c, avl.c, pqueue.c, main.c) with the benchmark results and chart above, generated from real program runs
- GitHub Upload — push the source files and this report to a repository and submit the link

Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5

CO–PO–Assessment Mapping

Assessment Component	CO(s)	PO(s)	Bloom's Level	Marks
Problem formulation			L3/L4	10
Application of knowledge			L3/L4	20
Solution / Design / Implementation			L4/L5/L6	20
Modern tool usage			L3/L4	10
Validation			L4/L5	15
Analysis & justification			L4/L5	15
Broader considerations			L4/L5	5
Documentation & reflection			L3/L4	5

Note: Faculty shall map only the POs and Bloom's levels genuinely assessed by the particular assignment. Every assignment is not required to map to every PO.

Faculty Evaluation & Continuous Improvement CO Attainment

Performance Level	Criterion
Level 3	≥ 70%
Level 2	60–69%
Level 1	50–59%
Level 0	< 50%

Target CO Attainment: _____

Actual CO Attainment: _____

Faculty Analysis

Areas in which students performed well: _____

Areas requiring improvement: _____

Corrective / Improvement Action: _____

Action to be implemented in the next offering: _____